

6 A cylindrical copper wire P of length 0.24 m is shown in Fig. 6.1.

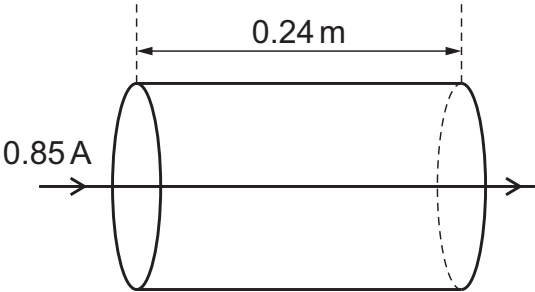


Fig. 6.1 (not to scale)

The current in the wire is 0.85 A.
The resistance of the wire is 3.3 mΩ.
The **total** number of charge carriers N in the wire is 2.6×10^{22} .
The resistivity of copper is $1.8 \times 10^{-8} \Omega \text{ m}$.

(a) Calculate the potential difference between the two ends of the wire.

potential difference = V [2]

(b) (i) Show that the cross-sectional area of the wire is $1.3 \times 10^{-6} \text{ m}^2$.

[2]

(ii) Show that the number density of charge carriers in the wire is $8.3 \times 10^{28} \text{ m}^{-3}$.

[1]

(iii) Calculate the average drift speed of the charge carriers (electrons) in the wire.

average drift speed = ms^{-1} [2]

(c) A different copper wire Q has the same volume as wire P, but non-uniform radius, as shown in Fig. 6.2.

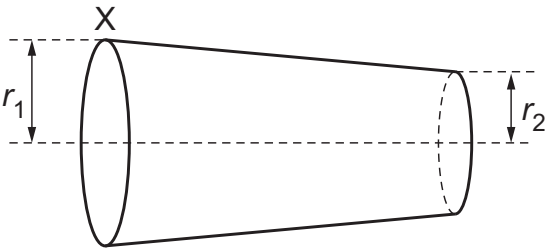


Fig. 6.2 (not to scale)

The radius r_1 at end X of wire Q is the same as the radius of wire P. Radius r_2 is less than r_1 .

(i) State and explain how the resistance of wire Q compares with the resistance of wire P.

.....
.....
.....
.....
.....
.....
..... [4]

(ii) On Fig. 6.3, sketch a graph of the variation of the average drift speed of the charge carriers with distance from end X of wire Q.

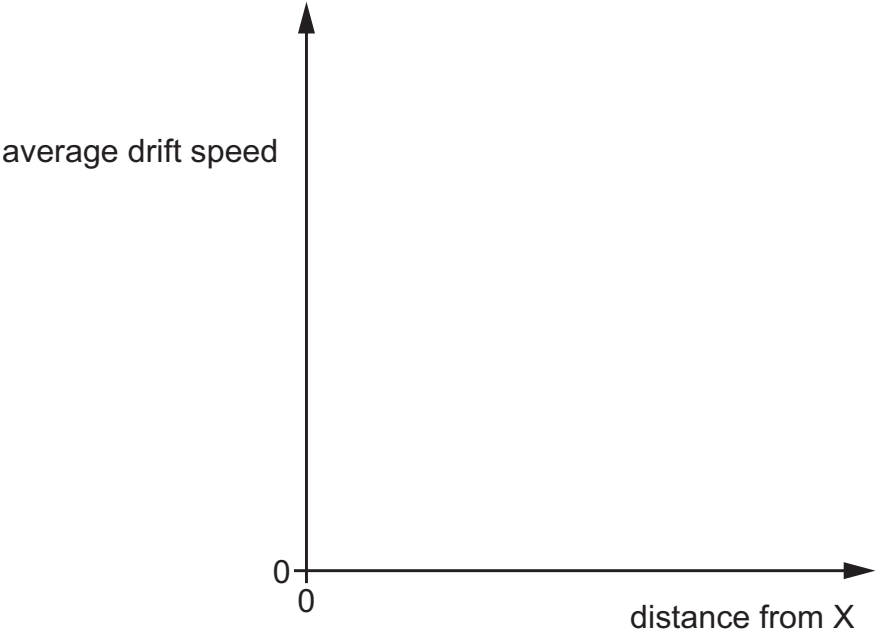


Fig. 6.3

[2]